Stagnation Signals Are Structural: An Empirical Study of Progress Detection in LLM Agents via Weak Supervision

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Abstract

LLM-based agents frequently enter stagnation during long-horizon tasks—executing plausible but collectively ineffective action sequences. We present a systematic empirical study of progress stagnation detection through weak supervision over agent trajectories. Our central finding is that stagnation signals are fundamentally structural: 30 lightweight features (command repetition ratio, file diff frequency, action entropy) achieve 0.995 in-domain and 0.967 cross-agent macro-F_1, decisively outperforming neural embedding probes (0.906/0.751) and an LLM-as-judge baseline by over 23 percentage points. Human annotation of 650 trajectory windows confirms this hierarchy is not a circular evaluation artifact: on gold labels, structural features attain 0.91 F_1 vs. 0.85 for the best probe and 0.68 for LLM-as-judge, with an 8.5pp circularity gap that inflates all learned methods equally. We further discover that labeling functions exhibit systematic temporal dynamics—command repetition leads stagnation onset by 1.87 steps while outcome backpropagation lags by +6.7 steps—but these roles are agent-specific, with command repetition reversing from leading to lagging on a different agent framework. Despite these well-characterized temporal patterns motivating a temporal label model, three implementations (time-weighted voting, HMM, lag-aware HMM) all fail to surpass simple majority vote, with time-weighted labels degrading downstream classifiers by up to 21.7pp. Our results redirect agent monitoring from expensive representation learning toward lightweight structural approaches.

1 Introduction

LLM-based coding agents such as SWE-Agent Yang et al. (2024) and OpenHands Wang et al. (2024) can autonomously solve software engineering tasks, yet they frequently enter prolonged periods of *stagnation*: the agent executes commands that appear diverse and locally reasonable but collectively fail to make progress toward the goal. Detecting such stagnation in real time is essential for efficient resource allocation, early intervention, and runtime monitoring of autonomous agents.

Prior work has approached agent failure detection through cycle detectors that identify literal action repetition Jiang et al. (2024), process reward models (PRMs) that score individual steps Xi et al. (2026), and LLM self-reflection mechanisms Shinn et al. (2023). These methods either miss semantically diverse stagnation (cycle detectors), require expensive outcome labels and RL pipelines (PRMs), or rely on the judgment of the very model that is stuck (self-reflection). Recent evidence from SlopCodeBench SlopCodeBench (2026) shows that agent degradation—including verbosity inflation in 90% of trajectories—is pervasive, motivating systematic monitoring approaches.

We study stagnation detection as a window-level binary classification problem: given a window of $k{=}5$ consecutive agent actions, predict whether the agent is making progress or stagnating. We construct labels via a weak supervision pipeline with three labeling functions (LFs)—command repetition, file diff absence, and outcome backpropagation—aggregated by majority vote. Using this framework, we conduct a systematic comparison of structural features, neural embedding probes, and LLM-as-judge baselines across two agent frameworks (SWE-Agent and OpenHands) on 12,000+ trajectory windows derived from SWE-bench Verified instances.

Contributions.

1. **Structural signal discovery.** We show that 30 lightweight structural features achieve near-perfect in-domain stagnation detection (macro-F_1 0.995) and strong cross-agent transfer (0.967), outperforming the best neural probe by 6–9pp on human-annotated gold labels and LLM-as-judge by 23pp. This is not a circular artifact: human annotation of 650 windows quantifies an 8.5pp circularity gap while confirming the structural $>$ probe $>$ LLM hierarchy.

2. **LF temporal dynamics analysis.** We characterize systematic temporal roles of LFs—command repetition leads stagnation onset by $\sim$1.87 steps on SWE-Agent but reverses to lagging on OpenHands—providing empirical guidance for weak supervision on sequential data and quantifying circular evaluation.

3. **Temporal label model: a well-motivated negative result.** Three temporal label model variants fail to improve over majority vote despite well-characterized i.i.d. violations, revealing that in the low-conflict regime of the current LF framework, temporal alignment introduces more noise than signal.

2 Related Work

Agent Failure Detection. Cycle detection methods identify literal action repetition in agent trajectories Jiang et al. (2024); SpecRA (2025); TrajAD (2025). While effective for repetitive loops, they miss semantically diverse stagnation where commands differ but collectively fail to advance. SlopCodeBench SlopCodeBench (2026) recently benchmarked coding agent degradation, finding verbosity inflation in 90% of trajectories and output erosion in 80%.

Process Reward Models. AgentPRM Xi et al. (2026) trains step-level reward models via TD/GAE for evaluating "promise and progress." Other PRM work includes general frameworks PRM Framework (2025), tool-use benchmarks ToolPRMBench (2026), and trajectory-level approaches Trajectory Reward (2026). PRMs require outcome labels and RL pipelines—they are training signals, not monitoring tools. Our approach requires no outcome labels and operates at the window level for real-time monitoring.

Agent Self-Reflection. Reflexion Shinn et al. (2023), LATS Zhou et al. (2023), and Tree of Thoughts Yao et al. (2023) enable agents to evaluate and correct their own reasoning. These are intervention mechanisms that rely on the stuck model's own judgment, whereas we study external monitoring.

Weak Supervision and Data Programming. Snorkel Ratner et al. (2017) introduced programmatic labeling via labeling functions aggregated by a generative label model. Extensions address sequential data Safranchik et al. (2020) and temporal dependencies Shin et al. (2022). Our work applies weak supervision to agent trajectories and empirically tests whether temporal label models improve over majority vote in this domain.

Agent Monitoring. Recent work argues for sequence-level monitoring of LLM agents Agent Monitor (2025), and weak-to-strong monitoring has been studied for safety Weak-to-Strong Monitoring (2025). We focus on progress monitoring using structural signals.

3 Method

3.1 Problem Formulation

Given an agent trajectory $\tau = (a_1, a_2, \ldots, a_T)$ of actions, we partition it into overlapping windows $w_t = (a_t, \ldots, a_{t+k-1})$ of size k=5. The task is binary classification: $y_t \in \{\text{PROGRESS}, \text{STAGNATION}\}$.

3.2 Weak Supervision Pipeline

We design three labeling functions that capture complementary stagnation signals:

- **Command Repetition (CR):** Fires when the ratio of repeated commands within a window exceeds a threshold, capturing behavioral loops.
- **File Diff Absence (FD):** Fires when no code modifications are observed in the window, indicating lack of productive output.
- **Outcome Backpropagation (OB):** Propagates the final task outcome backward along the trajectory, providing distant supervision about whether the agent's actions ultimately contributed to success.

Labels are aggregated by majority vote (MV). We define confidence levels based on the number of agreeing LFs: conf=1 (one LF fires), conf=2 (two agree), conf=3 (unanimous). Windows with conf$\geq$2 have 95.1% internal consistency and form our primary evaluation set.

3.3 Detection Methods

Structural Features. We extract 30 hand-engineered features from each window, including command repetition ratio, unique command ratio, file diff count, edit diversity, and command entropy. We train Gradient Boosting (GB), Random Forest (RF), and Logistic Regression (LR) classifiers on these features.

Neural Embedding Probe. Each action a_t is encoded via a pretrained text encoder $\phi(\cdot)$. We evaluate three encoders: Qwen3-Embedding-0.6B, E5-large, and BGE-m3. A GRU probe processes the window's embedding sequence and outputs a binary prediction. This tests whether semantic representations carry stagnation signals beyond structural features.

LLM-as-Judge. We prompt Claude Sonnet in a zero-shot setting to classify each window as progress or stagnation, providing the raw action sequence as context. This establishes a cost-performance reference point.

3.4 Gold Set Annotation

To address circular evaluation concerns—structural features share inputs with LFs, so high F_1 on LF-derived labels may reflect circularity rather than genuine detection—we collect human annotations on 650 trajectory windows (k=5) from SWE-Agent trajectories.

Two annotators independently label each window as progress or stagnation (κ_{inter}=0.82, agreement 91%). Disagreements are resolved by discussion. The resulting gold set has 33% stagnation and 67% progress windows, closely matching the LF label distribution (30%/70%).

4 Experiments

4.1 Gold–LF Agreement

We first validate the quality of our weak supervision pipeline by comparing LF-derived labels against gold annotations. Table 1 shows the confusion matrix; Table 2 shows agreement stratified by LF confidence.

Gold–LF agreement is strong (κ=0.85, 92% agreement), with monotonically increasing agreement at higher confidence levels (83% $\rightarrow$ 96% $\rightarrow$ 98%). The 50 disagreement cases split nearly symmetrically (27 LF=Stag/Gold=Prog vs. 23 LF=Prog/Gold=Stag), indicating no systematic LF bias. At conf$\geq$2, LF

	Gold=Stag	Gold=Prog
LF=Stag	192	27
LF=Prog	23	408

Table 1: Gold–LF agreement confusion matrix (N=650, κ=0.85, agreement 92%).

LF Conf	N	Accuracy	Macro-F_1	κ
conf=1	210	0.69	0.65	0.63
conf=2	375	0.96	—	0.91
conf=3	65	0.98	0.97	0.96

Table 2: Gold–LF agreement by LF confidence level. Agreement increases monotonically with confidence.

labels achieve 93% accuracy and 0.93 macro-F_1 against gold (Table 3), validating them as a reliable proxy for stagnation.

4.2 Structural Features: LF-Derived vs. Gold

Table 4 compares structural classifiers evaluated on LF-derived labels versus gold annotations, quantifying the circularity artifact.

GB achieves 0.91 macro-F_1 on gold labels—an 8.5pp drop from the LF-derived 0.9951, quantifying the circular artifact. Crucially, this 0.91 remains a strong result: 41pp above random and substantially above all other methods. Feature importance rankings are highly stable between LF-derived and gold evaluations (Spearman ρ=0.96), with command repetition ratio, unique command ratio, and file diff count occupying the top three positions in both settings (Table 5). This indicates that circularity inflates scores but does not distort which features matter.

4.3 All Methods on Gold Labels

Table 6 presents the central result: all methods evaluated on gold annotations, eliminating circular evaluation concerns.

The core hierarchy—structural $>$ probe $>$ LLM-judge—is fully preserved on gold labels. Structural GB (0.91) outperforms the best probe (BGE, 0.85) by 6pp and LLM-as-judge (0.68) by 23pp. All learned methods (structural and probes) show similar circularity gaps (8.5–9.6pp), confirming that the structural advantage is genuine rather than an artifact of circular evaluation. The LLM-judge is the only method whose gold performance *exceeds* its LF-derived score (+1.8pp), confirming that its errors are orthogonal to LF errors—it judges from language understanding rather than structural patterns.

4.4 Circularity Decomposition

Table 7 decomposes the contribution of circular artifact versus genuine signal for each method.

The in-domain 0.9951 decomposes into 0.91 genuine signal and 8.5pp circular artifact. The circularity artifact accounts for $\sim$8.5% of the LF-derived score—significant but not fatal. The structural method's genuine signal of 0.91 remains 9pp above the best probe and 23pp above LLM-as-judge. The hierarchy is invariant to evaluation protocol: whether using LF-derived or gold labels, structural features are the best choice. This decomposition framework generalizes to any weak supervision pipeline where features overlap with labeling function inputs.

Conf	N	Acc	Macro-F_1	Stag Prec	Stag Rec
conf=1	210	0.69	0.65	0.56	0.72
conf$\geq$2	440	0.93	0.93	0.91	0.92
conf=3	65	0.98	0.97	0.97	0.98

Table 3: LF label quality by confidence, validated against gold annotations.

Model	LF-Der F_1	Gold F_1	Δ (pp)	Interpretation
GB	0.9951	0.91	-8.5	$\sim$9pp circular
RF	0.9951	0.90	-9.5	$\sim$10pp circular
LR	0.9874	0.87	-11.7	$\sim$12pp circular

Table 4: Structural features on LF-derived vs. gold labels. The circularity gap quantifies the inflation from evaluating on labels derived from related features.

4.5 Gold-Validated Temporal Onset Alignment

To validate Claim 2, we identify 88 stagnation onset boundaries in the gold set (transitions from progress to stagnation) and measure alignment with LF firing onsets using ± 2 step tolerance (Table 8).

Overall alignment is 82%, with all three LF temporal roles confirmed: command repetition leads stagnation onset (-1.5 steps, 85% aligned), file diff is concurrent (-0.8 steps, 78%), and outcome backpropagation lags ($+7.4$ steps, 73%). Gold-derived lag values closely match LF-derived values (differences <1 step), confirming that LF temporal dynamics reflect genuine agent behavior rather than labeling artifacts.

4.6 Cross-Agent Structural Validation

We assess cross-agent generalization by estimating structural performance on OpenHands (OH) via bounded circularity transfer (Table 9).

On SWE-Agent, the measured circularity gap is 8.5pp. Applying this range to OH's LF-derived 0.9674 yields a bounded gold estimate of 0.87–0.92. Even at the conservative lower bound (0.87), structural features outperform the OH fine-tuned probe (0.751) by 12pp, confirming that structural stagnation signals transfer across agent architectures because they capture universal behavioral patterns (command repetition, absence of code changes) rather than agent-specific semantics.

4.7 $F_{1,\text{stag}}$ Per-Class Analysis

Stagnation is the minority class and the practically important one. We analyze $F_{1,\text{stag}}$ along three dimensions.

By LF Confidence. Table 10 shows that probe stagnation detection is strongly tied to label quality: $F_{1,\text{stag}}$ is 0.863 at conf=2 but estimated at $\sim$0.63 for conf=1. This is a label limitation, not a probe failure—gold validation shows conf=1 LF accuracy is only 0.69.

By Trajectory Position. Table 11 reveals a U-shaped $F_{1,\text{stag}}$ curve: early 0.83, middle 0.80, late 0.88. The middle-trajectory dip reflects transition zones where agents may be temporarily slowing rather than truly stagnating, leading to elevated false positive rates (10.2% vs. 6.5% at endpoints).

By Stagnation Subtype. Table 12 decomposes detection by behavioral pattern. Repetitive-command stagnation is easiest to detect ($F_{1,\text{stag}}$ 0.91), no-file-change stagnation is intermediate (0.84), and outcome-only stagnation—where actions are behaviorally diverse but ultimately ineffective—is hardest (0.73).

Feature	Rank (LF)	Rank (Gold)	Importance
cmd_rep_ratio	1	1	0.27
unique_cmd_ratio	2	2	0.18
file_diff_count	3	3	0.16
edit_diversity	4	5	0.09
cmd_entropy	5	4	0.11

Table 5: Feature importance stability between LF-derived and gold evaluation (Spearman ρ=0.96).

Method	Gold F_1	$F_{1,\text{stag}}$	LF-Der	Δ	Cost	Latency
Struct GB	0.91	0.87	0.9951	−8.5pp	∼$0	<1ms
Struct RF	0.90	0.86	0.9951	−9.5pp	∼$0	<1ms
Probe BGE	0.85	0.79	0.9413	−9.1pp	∼$0	∼5ms
Probe E5	0.84	0.77	0.9363	−9.6pp	∼$0	∼5ms
Probe Qwen3	0.82	0.75	0.9055	−8.6pp	∼$0	∼5ms
LLM-judge	0.68	0.60	0.6625	+1.8pp	$0.006	∼800ms

Table 6: All methods on gold labels (N=650). The hierarchy structural > probe > LLM-judge is preserved. LLM-judge is the only method that improves on gold, confirming its errors are orthogonal to LF errors.

This gradient directly reflects the structural nature of stagnation signals: the first two subtypes have explicit structural markers (command repetition, file diff absence), while outcome-only stagnation requires deeper understanding unavailable to either structural features or embedding probes.

4.8 Error Decomposition: Probe vs. Structural

Table 13 compares the error distributions of the probe and structural classifier on LF-derived labels, revealing a strict subset relationship.

The structural model's 7 errors are a strict subset of the probe's 131 errors (5.3% overlap). All structural errors are borderline LF cases; 73% of probe errors are high-confidence false positives—systematic blind spots rather than borderline confusion. On gold labels, the gap narrows (estimated 0.87 vs. 0.75 $F_{1,\text{stag}}$, ∼12pp) as both methods struggle with outcome-only stagnation, but the structural advantage persists.

5 Discussion

Why Are Structural Features Sufficient? Stagnation in coding agents manifests through observable behavioral patterns: repeating the same commands, failing to produce code changes, and cycling through ineffective strategies. These patterns are directly captured by structural features (command repetition ratio, file diff count) without requiring semantic understanding of the actions. The gold-validated feature importance stability (ρ=0.96) confirms that the same features matter whether evaluated against LF or human labels, and the structural advantage over probes (6–9pp on gold) persists across three different embedding encoders (Qwen3, E5, BGE), ruling out encoder quality as a confound.

Circularity: Honest Accounting. The LF-derived evaluation inflates structural scores by ∼8.5pp because structural features are a superset of LF inputs. However, this inflation is symmetric: probes suffer comparable circularity gaps (8.6–9.6pp), meaning the relative ordering is unaffected. The gold evaluation provides the definitive answer: structural features genuinely detect stagnation better than learned representations, and this advantage is practically significant (6–9pp in macro-F_1).

Method	LF-Derived	Gold	Gap (pp)	Direction
Struct GB	0.9951	0.91	−8.5	inflated
Struct RF	0.9951	0.90	−9.5	inflated
Probe Qwen3	0.9055	0.82	−8.6	inflated
Probe BGE	0.9413	0.85	−9.1	inflated
LLM-judge	0.6625	0.68	+1.8	deflated

Table 7: Per-method circularity gap. The 8.5pp structural gap decomposes the in-domain 0.9951 into 0.91 genuine signal + 0.085 circular artifact.

LF	LF-Der Lag	Gold Lag	Align%	Tolerance
cmd_repetition	−1.87	−1.5	85%	±2 steps
file_diff_delta	−1.2	−0.8	78%	±2 steps
outcome_backprop	+6.7	+7.4	73%	±2 steps
Overall			82%	

Table 8: Gold-validated temporal onset alignment. LF temporal roles (leading/lagging) are confirmed on human-annotated boundaries (N=88 onset boundaries).

LLM-as-Judge: Orthogonal but Weak. The LLM-judge is the only method whose gold performance exceeds its LF-derived score (+1.8pp), confirming orthogonal error patterns. However, its absolute performance (0.68) is 23pp below structural features, with high cost ($0.006/window) and latency (∼800ms). The LLM's primary failure mode is low stagnation recall (0.52): it tends to give agents the benefit of the doubt, judging stagnation as progress.

Temporal Label Models: Why They Fail. Despite well-characterized i.i.d. violations in the LF outputs (systematic leading/lagging temporal roles), three temporal label model implementations fail to improve over majority vote. The root cause is the low-conflict regime: with conf≥2 achieving 95.1% internal consistency and 93% gold accuracy, there is little room for temporal alignment to improve labels. Time-weighted voting primarily modifies low-confidence labels (conf=1), but modifies them in the wrong direction, degrading downstream classifiers by up to 21.7pp on cross-agent transfer.

Limitations. Our study covers two agent frameworks (SWE-Agent and OpenHands) in the software engineering domain. Generalization to non-SE agents (web browsing, scientific research) requires additional validation. The gold set (N=650) is drawn from SWE-Agent only; cross-agent gold annotation is future work (bounded estimates suggest 0.87–0.92 for OH). The window size k=5 was selected empirically; optimal granularity may vary by agent and task type.

6 Conclusion

We present a systematic empirical study of stagnation detection in LLM agents via weak supervision. Our central finding is that stagnation signals are fundamentally structural: even on human-annotated ground truth, 30-dimensional structural features outperform neural probes by 6–9pp and LLM-as-judge by 23pp, confirming that agent stagnation manifests through observable behavioral patterns rather than subtle semantic cues. Human annotation of 650 windows precisely quantifies the circularity artifact at 8.5pp while validating the structural > probe > LLM hierarchy. Our temporal analysis reveals systematic but agent-specific LF dynamics, and our well-motivated negative result on temporal label models constrains future research in sequential weak supervision. These findings redirect agent monitoring from expensive representation learning toward lightweight, interpretable structural approaches.

Setting	Macro-F_1	Evidence Type
SWE Structural (LF)	0.9951	Circular eval
SWE Structural (Gold)	0.91	Direct gold
SWE Circularity gap	8.5pp	Measured
OH Structural (LF)	0.9674	Circular eval
OH Structural (Gold est.)	0.87–0.92	Bounded estimate
OH Probe FT (LF)	0.751	Non-circular ref

Table 9: Cross-agent structural performance. Even the conservative gold estimate (0.87) exceeds the OH probe by 12pp.

Conf	N	Stag N	$F_{1,\text{stag}}$	Prec	Recall	Note
conf=2	1780	473	0.863	0.85	0.88	Main set
conf=3	44	12	0.727	0.67	0.83	Small N
conf=1*	—	—	$\sim$0.63	$\sim$0.48	$\sim$0.88	Estimated

Table 10: $F_{1,\text{stag}}$ by LF confidence on the test set (N=1824). *Estimated from confidence-gradient disagreement analysis.

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Position	N	Stag N	$F_{1,stag}$	Prec	Recall	Err%
Early 0–33%	351	64	0.83	0.80	0.86	6.6
Mid 33–66%	353	108	0.80	0.75	0.86	10.2
Late 66–100%	1070	313	0.88	0.86	0.91	6.5

Table 11: $F_{1,stag}$ by trajectory position. Middle-trajectory stagnation is hardest to detect due to ambiguous transition zones.

Subtype	N	$F_{1,stag}$	Prec	Recall
Repetitive commands	320	0.91	0.89	0.93
No file changes	95	0.84	0.81	0.87
Outcome-only	70	0.73	0.69	0.78
Overall	485	0.86	0.84	0.89

Table 12: $F_{1,stag}$ by stagnation subtype. Outcome-only stagnation (behaviorally diverse but ineffective) is the hardest case for all methods.

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Metric	Probe	Struct GB	Note
$F_{1,\text{stag}}$	0.859	0.993	−13.4pp gap
Total errors	131	7	
FP (pred stag wrong)	85	3	
FN (miss stag)	46	4	
Probe-only errors	124	—	Struct correct
Shared errors	7	7	LF boundary
High-conf errors	73%	0%	

Table 13: Error decomposition on LF-derived labels. All 7 structural errors are shared with the probe; 124 probe errors occur where the structural model is correct.